- Homework 5

Determine the minimum-time continuous-thrust transfer from an initial orbit of radius 10000 Km to a final orbit of radius 42164 km (GEO)

(using the canonical units DU and TU in integrations)

with the following propulsion parameters

$$c = 30\ \frac{\text{Km}}{\text{sec}} \qquad \text{and} \qquad u_T^{(max)} = 0.01\ g_0$$

Local search on $\{\lambda_{10}, \lambda_{30}, \lambda_{40}, t_f\}$ can be performed, for instance using fmincon in Matlab, starting from the following guess values

$$\lambda_{10}^{(G)} = -1$$

$$\lambda_{30}^{(G)} = 0.2$$

$$\lambda_{40}^{(G)} = -1$$

$$t_f^{(G)} = 40 \ (\text{TU})$$

Plot the time histories of

(i) radius

(ii) velocity components $v_r, v_t$

(iii) thrust pointing angle

(iv) transfer trajectory

```matlab
%% MAIN FILE

guess=[0.2,-1,-1,10]; % converging guess

% accessory matrices for fmincon
A=[]; B=[]; Aeq=[]; Beq=[];
LB=[]; UB=[];

% OPTION 1: use minTimeTransf.m and nonlinconstr.m
[xopt,optval,outc]=fmincon('minTimeTransf',guess,A,B,Aeq,Beq,LB,UB,'nonlinconstr');

% OPTION 2: use minTimeTransf2.m
[xopt,optval,outc]=fminsearch('minTimeTransf2',guess);

%%---------------------------- OPTION 1 -----------------------------------
function minTime=minTimeTransf(xx);
% function that accepts the unknown quantities (lam10,lam30,lam40,tf) and
% returns the value of the objective function
% [...]

function [ineq,equal]=constraint(xx);
% function that accepts the unknown quantities (lam10,lam30,lam40,tf) and
% returns ineq = vector of the inequality constraints
%         equal = vector of the equality constraints
% [...]

%%---------------------------- OPTION 2 -----------------------------------
function bndConds=minTimeTransf2(xx);
% function that accepts the unknown quantities (lam10,lam30,lam40,tf) and
% returns the value of magnitude of the boundary condition violations
% [...]
```